

Naveed Ahmed

Principal Software Engineer | AI Solution Architect

Karachi, Sindh, Pakistan | +92 332 3919193 | nksherani@outlook.com

linkedin.com/in/naveed-ahmed-96b45552 | Portfolio

PROFESSIONAL SUMMARY

Principal Software Engineer and AI Solution Architect with 9+ years of experience in software architecture, including 7 years of deep expertise in Microsoft Azure cloud services and 1.5 years of specialized, hands-on experience in AI and MLOps. Proven ability to translate complex business problems into practical AI solutions, leading end-to-end delivery from data preparation and feature engineering to model training, deployment, and production monitoring. Deeply experienced in Generative AI, Agentic AI, RAG, and LLMs (including Local LLMs). Highly proficient in creating RFP responses, technical roadmaps, and engaging with technical and executive stakeholders to drive AI adoption across diverse domains.

SKILLS

1. Generative AI, LLMs & Agentic AI

- **Agentic AI Workflows:** Hands-on experience designing and building multi-agent systems using **LangGraph** (specifically implementing cooperative Content, Engagement, Strategy, and Optimization agents) with brand-safety guardrails.
- **RAG (Retrieval-Augmented Generation):** Expert in designing enterprise RAG architectures using **OpenAI embedding models**, specialized context-preserving chunking strategies, and custom high-fidelity page/clause citation mapping systems.
- **Local & Cloud LLMs:** Experience hosting, orchestrating, and prompting Local LLMs and commercial APIs (GPT-4, Gemini, GLM) for text generation, semantic reasoning, and database interfacing.
- **Prompt Engineering & Tool Calling:** Designing structured multi-step prompts (single-prompt categorizations, binary cascades, and hybrid majority-voting workflows) and integrating native model tool calling.

2. Data Science & Traditional AI/ML

- **Model Selection & Training:** Practical experience training, evaluating, and deploying traditional ML algorithms, including **XGBoost**, **Support Vector Classifiers (SVC)**, and **Linear Regression**.
- **Fine-Tuning:** Experienced in fine-tuning deep learning models (such as **BERT** and domain-specific LLMs) with custom loss functions and hyperparameters.
- **NLP Preprocessing & Feature Engineering:** Implementing text embeddings, data normalization, acronym resolution, synthetic data generation, and custom noise reduction.
- **Data Cleansing with AI:** Utilizing LLMs and AI tools to augment and denoise small or low-quality datasets, achieving significant accuracy gains (e.g., reaching 80% accuracy on corporate PII classification).

3. Cloud Architecture & Enterprise Integration (Azure Focus)

- **Azure AI & Cognitive Services:** In-depth implementation of **Azure OpenAI**, **Azure AI Search**, **Azure AI Foundry**, **Azure Language Service (NER)**, and **Azure Document Intelligence (DI)**.
- **Azure Data & Infrastructure:** Architecting enterprise-scale solutions using **Azure Blob Storage**, **Cosmos DB**, **Azure Service Bus**, **SQL Server**, **PostgreSQL**, **Redis**, and **API Management**.
- **Microservices & Orchestration:** 7+ years of experience migrating monolithic legacy architectures to microservices SaaS platforms using **Azure Kubernetes Service (AKS)** and Docker.
- **Distributed Data Pipelines:** Designing robust, real-time message integration systems using **Apache Kafka**, **Azure Service Bus**, and **AWS EventBridge** for high-throughput event processing.

4. MLOps, LLMOps & Automation

- **ML Workspace Orchestration:** Managing the full machine learning lifecycle within **Azure ML Workspace**.
- **Pipeline Automation:** Building fully automated pipelines spanning raw data ingestion (Azure Blob Storage) → automated preprocessing → model training and hyperparameter tuning → deployment of managed production endpoints.

- **AutoML & Model Selection:** Utilizing **Azure AutoML** to automate model selection based on performance metrics.
- **CI/CD & CLI Tooling:** Deeply proficient in **Azure CLI**, **Firebase CLI**, **AWS CLI**, and **GitHub CLI** to streamline infrastructure as code (IaC) and deployment pipelines.
- **Model Context Protocol (MCP):** Developing agentic database connections and workflows using the Model Context Protocol to translate natural language into business intelligence queries.

5. Traditional Software Development

- **Programming Languages & Backend Frameworks:** Strong delivery experience with **Python** (including **FastAPI**), **C# / .NET (6/8)**, **Node.js**, and **.NET Web API**.
- **Frontend & Mobile:** **JavaScript**, **TypeScript**, **React.js**, **Next.js**, and **React Native** (Android/iOS), including Firebase-backed consumer apps.
- **Application Foundations:** REST API design; **Azure AD SSO** and **Google Auth**; shared middlewares for logging and global exception handling; **Hangfire** background jobs; Domain-Driven Design (DDD) domain services.
- **Domains & Integrations:** Ecommerce/POCs, payment gateways, and financial software; multi-tenant SaaS (clients, teams, roles); third-party social and ads APIs (Meta, X, YouTube, LinkedIn, TikTok).

6. Stakeholder Management & Consulting

- **RFP & Proposal Preparation:** Experienced in preparing technical proposals, RFP responses, scope documents, and detailed effort estimation models.
- **Architecture Diagrams:** Designing clear, end-to-end system architectures across application, integration, and security layers.
- **Consulting & Leadership:** Proven ability to engage with business, technical, and C-level executive stakeholders to drive AI adoption across diverse domains (including auditing and fintech).

WORK EXPERIENCE

Principal Software Engineer I 10Pearls Pakistan

Karachi, PK (International Clients) | January 2025 – Present * **Technical Architecture & Strategy:** Lead technical architecture and engineering strategy for enterprise AI and cloud projects across international clients. * **AI Solution Design:** Architect enterprise RAG platforms, Agentic AI workflows, and conversational AI systems using Azure AI Search, Azure AI Foundry, Pinecone, and OpenAI APIs. * **End-to-End ML Ownership:** Translate complex domain requirements into technical architectures, overseeing data sources, data pipelines, and production deployment. * **MLOps & LLMops Pipeline Automation:** Build automated MLOps and LLMops pipelines in the Azure ML Workspace, managing the lifecycle from preprocessing triggers through to managed endpoint deployment. * **Data Science Initiatives:** Drive data science initiatives by producing data quality reports, refining feature engineering, and utilizing Azure AutoML.

Technical Lead I Turing.com (Contract)

Remote, US | July 2022 – January 2025 * **Client Healthcare Facility Platform (from scratch):** Owned solution creation for a US healthcare client's facility-management product—revamped the tech stack, stood up the core SaaS architecture and database design, and built/led the engineering team to deliver it. * **Legacy Integration & Continuity:** Took ownership of the legacy desktop codebase and initially integrated it with the new SaaS backend so operations stayed in sync while modernization progressed. * **Domain-Driven Architecture:** Broke the system into domain services following Domain-Driven Design, establishing clear bounded contexts for scheduling, time cards, shifts, HR, payroll, and coordination. * **Identity & Platform Foundations:** Implemented authentication with Azure AD SSO, plus shared middlewares for enriched logging and global exception handling. * **Background Processing:** Introduced Hangfire to run reliable background jobs across the platform. * **Cloud Delivery:** Designed application, integration, and infrastructure layers on Azure (App Service, AKS, Azure SQL).

Senior Software Engineer I VentureDive

Remote | September 2020 – April 2022 * **Multi-Client Delivery:** Worked as a contractor across multiple clients including Daraz, Motiv, and PayActiv, spanning ecommerce ideas/POCs, payment gateways, and financial software. * **Payments & Financial Systems:** Built and integrated payment gateway flows and financial software features for client platforms. * **Performance Optimization & Audit:** Drove performance optimization and technical audits to improve reliability, scalability, and operational health. * **Stack & Integrations:** Delivered solutions using .NET, Node.js, PostgreSQL, and event-driven architectures, including data pipelines and integrations with Kafka, Azure Service Bus, and AWS EventBridge.

Senior Software Engineer I Stingray Technologies (Pvt) Ltd

Karachi, PK | October 2017 – August 2020 * **Defence Equipment Software Systems:** Built and evolved software systems for defence equipment—started as a developer and grew into ownership of core system development. * **Real-Time Distributed Architecture:** Owned the core platform built on a distributed architecture that depended on efficient delivery of thousands of real-time messages under strict performance constraints. * **Simulators:** Developed and maintained simulators used to exercise, validate, and support the defence equipment systems. * **Stack & Delivery:** Delivered distributed, CPU-intensive systems and APIs using C#, SQL, and MongoDB.

Software Engineer I Techlogix

Karachi, PK | December 2016 – October 2017 * **Enterprise Scalability:** Developed highly scalable enterprise solutions for local and international clients, utilizing .NET Web API and SQL Server.

KEY PROJECTS PORTFOLIO

1. Tax Transaction Categorization Engine

Tagline: Automated Capital Allowances and Detailed Profit & Loss classification for enterprise auditing

Category: FinTech / Tax Compliance & Auditing (UK market client)

Platforms: Enterprise Cloud (Azure)

Tech Stack: Python, .NET, Azure App Service, React, Azure SQL Database, Azure ML Workspace, AutoML, BERT

*Detailed Profit & Loss (DPL) Classifier: Automatically categorizes high-volume DPL transactions into 76 distinct accounting and tax categories. Achieved 80% accuracy on classification using a refined, AI-cleansed dataset. **

*Traditional ML & Fine-Tuning Experimentation: Evaluated and experimented with multiple machine learning algorithms including XGBoost, Linear Regression, and SVC (Support Vector Classifier), alongside the fine-tuning of BERT models using different loss functions and hyperparameters. **

*Tangible Fixed Assets (TFA) Capital Allowances Classifier: Designed an LLM-based architecture using prompt engineering (single prompt, binary yes/no, and a hybrid consensus majority-voting framework) to classify capital additions into 6 Capital Allowances pools. **

Automated Data & MLOps Pipelines: Created the complete Azure infrastructure utilizing Azure ML Workspace to automate ingestion (Azure Blob Storage), raw preprocessing, training, metrics-based algorithm selection, and continuous deployment of the final model to a production endpoint.*

2. Enterprise Document Q&A & Citation Engine (RAG)

Tagline: Scalable Retrieval-Augmented Generation with custom document citation logic for audit compliance

Category: Enterprise AI / Knowledge Management & Auditing (UK market client)

Platforms: Enterprise Cloud (Azure)

Tech Stack: Python, React, Pinecone Vector DB, OpenAI Embedding Models & API

*Scalable Ingestion & Performance: Resolved major system timeouts and processing bottlenecks in a legacy RAG application, allowing it to efficiently handle large audit documentation without timing out. **

*Accurate Source Citations: Developed a custom citation mapping engine that links generated LLM answers back to exact pages, clauses, and reference text within the original document—a critical compliance capability not supported out of the box. **

Architectural Redesign & Optimization: Designed and implemented an alternative, scalable RAG architecture using Pinecone Vector Database and OpenAI embedding models, optimizing text chunking and overlap strategies to prevent context fragmentation during search.*

3. Custom Document Anonymization Tool & Data Pipeline

Tagline: Custom NLP pipeline for PII sanitization and redaction in financial auditing documents

Category: Enterprise AI / Data Privacy & Compliance

Platforms: Azure (Language Service / Named Entity Recognition / Document Intelligence)

Tech Stack: Python, FastAPI

*Sanitization & Redaction: Built a custom data anonymization pipeline and standalone tool utilizing Azure NER (Named Entity Recognition) and Azure Document Intelligence (DI) to identify and redact sensitive PII/audit data from raw PDFs, scans, and spreadsheets prior to pipeline ingestion. **

Redaction Logic: Developed rule-based and machine-learning-based redaction layers to resolve acronyms, custom enterprise identifiers, and context-specific sensitive terms.

4. Conversational Database Interface

Tagline: Agentic SQL/NoSQL interface translating natural language queries into business insights

Category: Enterprise AI / Conversational BI (Business Intelligence)

Platforms: Model Context Protocol (MCP) / Python / LLMs

Tech Stack: Python, React

*Agentic Workflows: Engineered custom agentic workflows utilizing Model Context Protocol (MCP) sidecars/servers to bridge LLMs with database connections. * Natural Language to Query Translation.** Implemented advanced prompt engineering and schema-context mapping to ensure reliable generation of complex database queries, providing explainable query construction and reasoning to users.

5. Agency Platform (Social Media Agency Automation SaaS)

Tagline: AI-operated social media management for agencies

Category: Business & Marketing SaaS

Platforms: Web (self-hosted, Docker)

Tech Stack: Python, React, PostgreSQL, LangGraph

*LangGraph AI Agents: Integrated AI agents powered by LangGraph for Content, Engagement, Strategy, and Optimization tasks, complete with a brand-safety guardrails layer. * Multi-Platform Integration.** Connected Meta Graph API, X API, Google YouTube API, LinkedIn API, and TikTok API, automating calendar publishing and a unified inbox with AI-drafted replies and human-in-the-loop approvals.

6. Healthcare Facility Management Platform

Tagline: Modernizing a legacy desktop facility-management system into APIs and mobile apps for scheduling, staffing, and HR

Category: Healthcare / Workforce & Facility Operations

Platforms: Microsoft Azure / Mobile (iOS & Android)

Tech Stack: .NET, C#, Azure, Azure SQL, Azure App Service, AKS, Hangfire, Azure AD SSO

*Modernization & Transition: Converted established facility-management business rules from a legacy desktop application into secure, scalable APIs on Azure. * Feature Expansion.** Added workforce scheduling, shift assignment, time cards, and mobile access for staff, powered by domain services (DDD), AKS orchestration, and Hangfire background jobs.

EDUCATION

Bachelor of Engineering (Computer Engineering)

NED University of Engineering and Technology | 2013 – 2016 * **Academic Performance:** CGPA 3.86 / 4.00